$$\sum_{i=1}^{k} i^3 + (k+1)^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$

Then substitute our n=k target into our summation

$$\left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$

$$\frac{k^2(k+1)^2}{4} + \frac{4(k+1)^3}{4} = \text{omitted}$$

$$\frac{\left[k^2 + 4(k+1)\right](k+1)^2}{4}$$

simplifying only the numerator

$k^2 + 4k + 4$

$(k+2)^2(k+1)^2$

$$\frac{(k+2)^2(k+1)^2}{4} = \frac{(k+1)^2(k+2)^2}{4}$$

✓